



AJKD Atlas of Renal Pathology: Glomerulonephritis With Dominant C3

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Clinical and Pathologic Features

Glomerulonephritis with dominant C3, a form of C3 glomerulopathy, typically presents with subnephrotic proteinuria and hematuria. Although a minority of patients may present with nephrotic syndrome, about half present with hypertension and impaired glomerular filtration rate. Up to 15% progress to end-stage renal disease. C3 glomerulopathy is rare with a typical age of onset of 30 years (range, 7-70 years).

Light microscopy: Most biopsies show a membranoproliferative pattern. Other less common histologic patterns include mesangial proliferative, diffuse endocapillary hypercellularity, and crescentic.

Immunofluorescence microscopy: Dominant or only C3 mesangial and capillary loop staining, at least 2+ stronger than immunoglobulin and C1q staining.

Electron microscopy: Mesangial and sub-endothelial deposits with occasional subepithelial deposits, possibly hump-like, without dense transformation of the deposits.

Etiology/Pathogenesis

Abnormalities in alternative complement pathway regulation, genetic or acquired.

Differential Diagnosis

Dense deposit disease can present with similar light microscopy and immunofluorescence findings, but has diagnostic ribbon-like dense deposits within the glomerular basement membrane, not seen in this form of C3 glomerulopathy. Postinfectious glomerulonephritis can present with a similar pattern of C3 dominant staining by immunofluorescence; however, endocapillary hypercellularity with numerous neutrophils and subepithelial hump-like deposits revealed by electron microscopy are features that help distinguish it from glomerulonephritis with dominant C3.

Some patients with pathologic features indistinguishable from typical postinfectious glomerulonephritis have an atypical prolonged clinical course, and may have underlying complement dysregulation, indicative of a C3 glomerulopathy.

Key Diagnostic Features

- Only C3 or dominant C3 staining, 2+ greater than immunoglobulin or C1q shown by immunofluorescence
- No dense transformation of the glomerular basement membranes revealed by electron microscopy

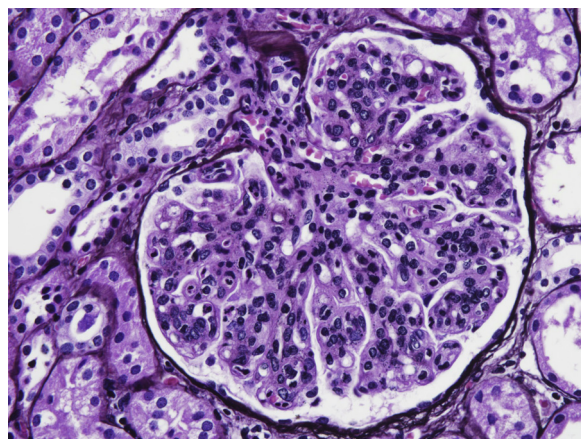


Figure 1. Glomerulonephritis with dominant C3 demonstrating endocapillary hypercellularity (Jones silver stain).

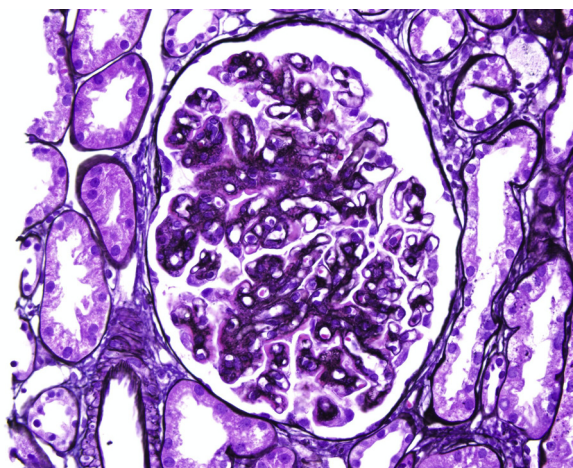


Figure 2. Glomerulonephritis with dominant C3 showing contour duplication of glomerular basement membranes (Jones silver stain).

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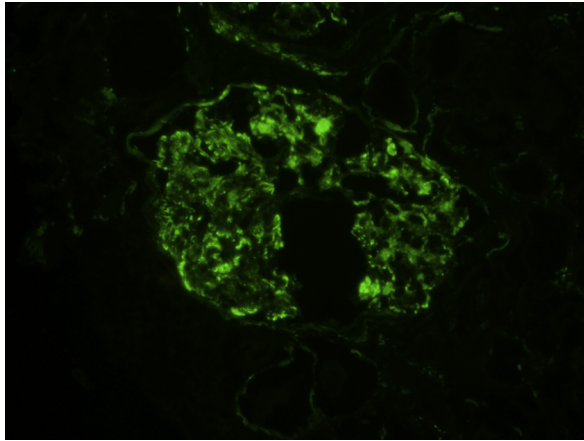


Figure 3. Glomerulonephritis with dominant C3, with strong mesangial and capillary wall C3 staining (immunofluorescence microscopy, C3).

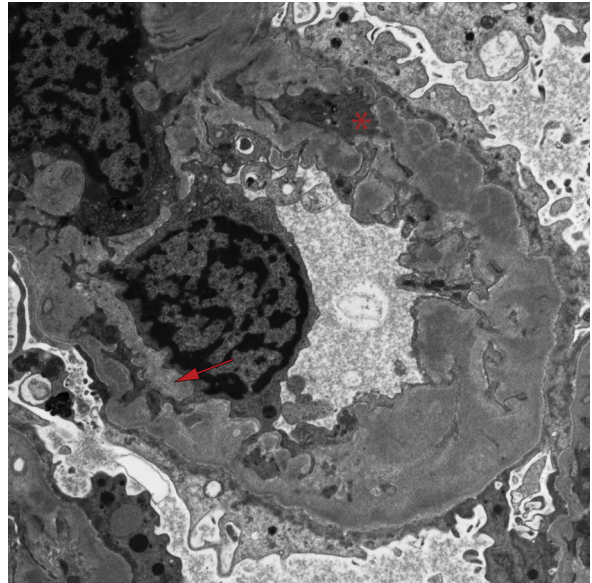


Figure 4. Glomerulonephritis with dominant C3, with subendothelial deposits associated with interposed cell (asterisk) and new basement membrane material between the interposed cell and endothelium (arrow; electron microscopy).